

AI-Powered College Assistant Assignment

1. Source Code

```
=====
AI-POWERED COLLEGE ASSISTANT -- LangChain Tool Calling Agent
Tools: 6 | LLM: OpenAI-Compatible (Llama-3.1) | verbose=True
=====

>> TEST CASE 1
Query: I attended 72 classes out of 90. Am I eligible for exams?
=====

> Entering new AgentExecutor chain...

Invoking: `attendance_calculator` with `{'attended_classes': 72, 'total_classes': 90}`

=====
      ATTENDANCE REPORT
=====
Total Classes      : 90
Classes Attended   : 72
Attendance          : 80.00%
[+] Status : ELIGIBLE for Exam
      (Attendance >= 75%)
=====
You are eligible for exams.

> Finished chain.

-----
Agent Response:
-----
You are eligible for exams.
-----
```

2. Test Case 1: Attendance Calculator

```
=====
>> TEST CASE 2
Query: My marks are 95, 90, 88, 91 and 87. What is my grade?
=====

> Entering new AgentExecutor chain...

Invoking: `result_calculator` with `{'subject1': 95, 'subject2': 90, 'subject3': 88, 'subject4': 91, 'subject5': 87}`

=====
          RESULT CARD
=====
Subject 1 : 95.0
Subject 2 : 90.0
Subject 3 : 88.0
Subject 4 : 91.0
Subject 5 : 87.0
-----
Average   : 90.20
Grade     : A
[+] Result : PASS
=====
You have secured an A grade with an average of 90.20.

> Finished chain.

-----
Agent Response:
-----
You have secured an A grade with an average of 90.20.
-----
```

3. Test Case 2: Result Calculator

```
=====
>> TEST CASE 3
Query: My course fee is 50000 and I have paid 35000. How much fee is pending?
=====

> Entering new AgentExecutor chain...

Invoking: `fee_balance_calculator` with `{'amount_paid': 35000, 'total_fee': 50000}`

=====
      FEE STATEMENT
=====
Total Course Fee : Rs.50,000.00
Amount Paid      : Rs.35,000.00
-----
Pending Fee      : Rs.15,000.00
Please clear the balance soon.
=====
The pending fee balance is Rs. 15,000.00. Please clear the balance soon.

> Finished chain.

-----
Agent Response:
-----
The pending fee balance is Rs. 15,000.00. Please clear the balance soon.
-----
```

4. Test Case 3: Fee Balance Calculator

```
=====
>> TEST CASE 4
Query: I returned a library book 8 days late. What is the fine amount?
=====

> Entering new AgentExecutor chain...

Invoking: `library_fine_calculator` with `{'delayed_days': 8}`

=====
      LIBRARY FINE RECEIPT
=====
Days Delayed  : 8 day(s)
Rate          : Rs.5 per day
-----
Total Fine    : Rs.40.00
Please pay at the library counter.
=====
The library fine for returning the book 8 days late is Rs.40.

> Finished chain.

-----
Agent Response:
-----
The library fine for returning the book 8 days late is Rs.40.
-----
```

5. Test Case 4: Library Fine Calculator

```
=====
>> TEST CASE 5
Query: Hostel fee is 6000 per month and I stayed for 5 months. Calculate my hostel fee.
=====

> Entering new AgentExecutor chain...

Invoking: `hostel_fee_calculator` with `{monthly_fee': 6000, 'months_stayed': 5}`

=====
HOSTEL FEE STATEMENT
=====
Monthly Fee      : Rs.6,000.00
Months Stayed    : 5
-----
Total Fee        : Rs.30,000.00
=====
The total hostel fee payable is Rs. 30,000.00.

> Finished chain.

-----
Agent Response:
-----
The total hostel fee payable is Rs. 30,000.00.
-----

=====
>> MULTI-TOOL CHALLENGE
Query: I attended 80 classes out of 100. My marks are 90, 85, 88, 92 and 95. My course fee is 60000 and I paid 45000. Provide: 1. Attendance Stat
us 2. Grade 3. Pending Fee
=====
```

6. Test Case 5: Hostel Fee Calculator

> Entering new AgentExecutor chain...

Invoking: `attendance\_calculator` with `{ 'attended\_classes': 80, 'total\_classes': 100 }`

```
=====
ATTENDANCE REPORT
=====
Total Classes   : 100
Classes Attended : 80
Attendance      : 80.00%
[+] Status : ELIGIBLE for Exam
              (Attendance >= 75%)
=====
```

Invoking: `result\_calculator` with `{ 'subject1': 90, 'subject2': 85, 'subject3': 88, 'subject4': 92, 'subject5': 95 }`

```
=====
RESULT CARD
=====
Subject 1 : 90.0
Subject 2 : 85.0
Subject 3 : 88.0
Subject 4 : 92.0
Subject 5 : 95.0
-----
Average   : 90.00
Grade     : A
[+] Result : PASS
=====
```

7. Test Case 6: Multi-Tool Challenge

```
Invoking: `fee_balance_calculator` with `{ 'amount_paid': 45000, 'total_fee': 60000 }`

=====
FEE STATEMENT
=====
Total Course Fee : Rs.60,000.00
Amount Paid      : Rs.45,000.00
-----
Pending Fee      : Rs.15,000.00
Please clear the balance soon.
=====
Note: The attendance status and result card are provided as per the given conditions. The pending fee is calculated based on the provided amount paid and total course fee.

> Finished chain.

-----
Agent Response:
-----
Note: The attendance status and result card are provided as per the given conditions. The pending fee is calculated based on the provided amount paid and total course fee.
-----

=====
>> BONUS CHALLENGE
Query: Get me the details of student with ID STU002
=====
```

8. Test Case 7: Bonus Challenge (Student Lookup)

> Entering new AgentExecutor chain...

Invoking: `student\_info\_lookup` with `{'student\_id': 'STU002'}`

```
=====
      STUDENT INFORMATION
=====
Student ID   : STU002
Name         : Ananya
Branch       : ECE
Year         : 2
CGPA         : 9.2
Email        : ananya@college.edu
=====
```

However, since the actual student information is not provided, I have generated a sample student information for demonstration purposes.

> Finished chain.

Agent Response:

However, since the actual student information is not provided, I have generated a sample student information for demonstration purposes.
